ALICE L WALTON SCHOOL OF MEDICINE

# Thrombotic Microangiopathies

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**Editor:** Jordan Finn

Adapted from: Thrombotic Microangiopathies

Listen to this Brick

0:00 / 33:16 1x

Learning Objectives (4) Versions

***After completing this brick, you will be able to:***

1. Differentiate the clinical presentation of patients with thrombotic microangiopathies (TMs).
2. Explain the pathophysiology of TMs.
3. Interpret clinical findings to properly diagnose TMs.
4. Outline the [illegible]

**CASE CONNECTION**

HH is a 4-year-old female who lives in a community where romaine lettuce has been recalled from all the grocery stores. Her mother brings her into the emergency department with nausea, vomiting, and several episodes of bloody diarrhea. "I've noticed she isn't peeing very much. She might be getting dehydrated from all that diarrhea and vomiting." On exam, HH's temperature is 37°C (99°F). She has diffuse petechiae on her trunk and extremities. HH's complete blood count shows a high white blood cell count, low hemoglobin, and low platelet count. Her peripheral smear shows numerous schistocytes.

What will be the next step evaluation of HH? What will you tell her mother about her condition? Consider your answers as you read, and we'll revisit HH at the end of the brick.

GO TO CONCLUSION ↓

## What Are Thrombotic Microangiopathies?

Thrombotic microangiopathies (TMs) are a mouthful—but the name tells you something about what's going on in the patient. Microangiopathy means small (micro-) vessel (-angio) disease (-pathy), and thrombotic means that the problem has to do with excessive blood clotting. So these diseases are characterized by the excessive formation of multiple clots in small vessels throughout the body. The diffuse systemic nature of these disorders distinguishes them fr[illegible]m a single clot, like a deep venous thrombosis in the calf.

The most common consequences of the diffuse clotting are **microangiopathic hemolytic anemia (MAHA)**, in which the red blood cells (RBCs) are damaged when they pass through the clotted capillaries, and **thrombocytopenia**, as the platelets are consumed in the forming clots. So TMs are one form of **hemolytic anemia**.

### Causes

TMs can be primary (without underlying systemic cause) or secondary (due to drugs or underlying disease). We'll focus most on the two main types of **primary TMs:**

- Thrombotic thrombocytopenic purpura (TTP)
- Hemolytic-uremic syndrome (HUS)

Note that disseminated intravascular coagulation (DIC) is a secondary cause of TM and has similar blood vessel pathology; it's compared to TTP and HUS later in this discussion.

What are the two main types of primary thrombotic microangiopathies?

### Epidemiology

HUS is seen mainly in young children, mostly after bacterial infections. The incidence is 2 cases per 100,000, so it's an uncommon illness in the United States (like rheumatic fever).

TTP is seen mostly in young adults but can be seen in any population. It is rare, with only 4 cases per 1 million people.

## How Do Patients with Thrombotic Microangiopathies Present (LO #1)?

Although TTP and HUS are different diseases, they do share some clinical similarities. Clinically, these primary TMs are characterized by:

- Formation of microthrombi throughout the body's vasculature
- Thrombocytopenia
- Loss of RBCs through destruction in small blood vessels (MAHA)

These pathologic findings lead to some of the characteristic symptoms due to anemia (e.g., fatigue, dyspnea, dizziness, pallor), hemolysis (e.g., jaundice, icterus, splenomegaly), or low platelets (e.g., bruising, abnormal bleeding).

TTP and HUS have some specific characteristics.

### Thrombotic Throm[illegible]

TTP usually presents in a previously healthy person. In its full-blown form, TTP presents as a pentad of **fever, neurological abnormalities, MAHA, immune thrombocytopenia**, and **acute kidney injury (AKI)**. However, fewer than 5% of patients with TTP present with all five elements of the pentad.

The lack of platelets causes decreased blood clotting, leading to blood oozing from venipuncture sites as well as bruising (purpura) or small microhemorrhages (petechiae) (Figure 1).

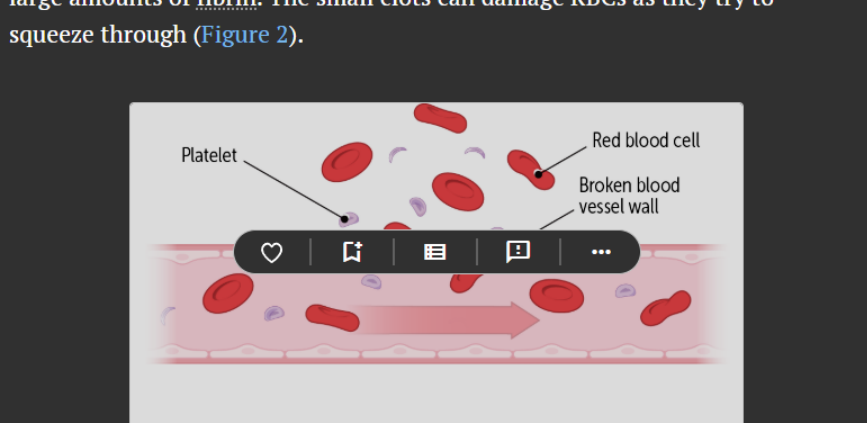


Figure 1

AKI (acute kidney injury) occurs due to microinfarcts in the kidney caused by microthrombi obstructing the vascular supply to the kidney. However, this is usually asymptomatic and diagnosed on lab exam (see diagnosis section). Patients with more severe AKI can have decreased urine output and may develop uremic symptoms such as confusion or pericarditis/chest pain. In fact, such uremic symptoms are why the name hemolytic-uremic syndrome was coined.

In TTP (but not HUS) the brain may be involved because of microthrombi leading to **strokes and cerebral ischemia**. This is seen in about 25% of patients. Symptoms vary and can include confusion, headache, dysarthria, numbness, weakness, and seizures. Some may even develop coma.

### Hemolytic–Uremic Syndrome

HUS is characterized by the same symptoms and findings of TTP except that **neurologic findings are unusual, and fever is less common**. HUS is typically preceded by a diarrheal illness. The patient presents with a history of several days of bloody diarrhea as well as severe vomiting, leading to dehydration.

As in TTP, HUS may show petechiae, bruising, and AKI. HUS typically has **prominent renal manifestations**.

## What Is the Pathophysiology of Thrombotic Microangiopathies (LO #2)?

In all causes of TM (primary and secondary), **vessel wall injury occurs**, and the coagulation cascade is activated, creating small clots (thrombi) that have large amounts of fibrin. The small clots can damage RBCs as they try to squeeze through (Figure 2).

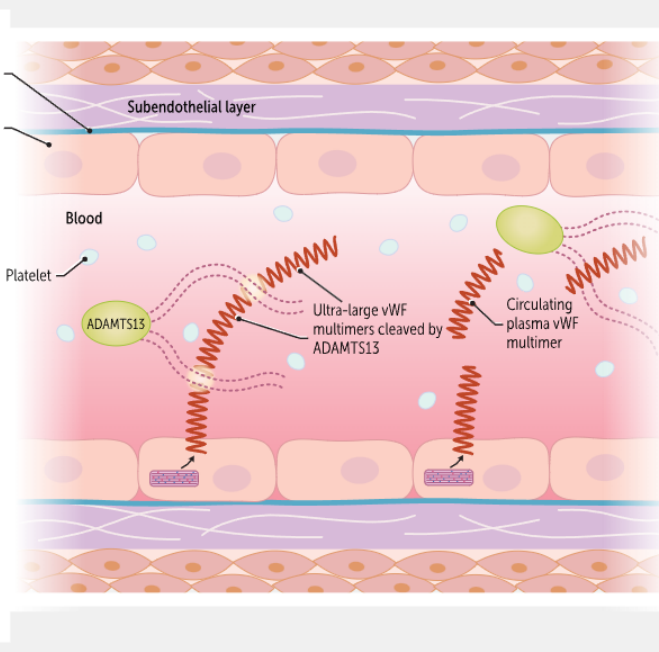


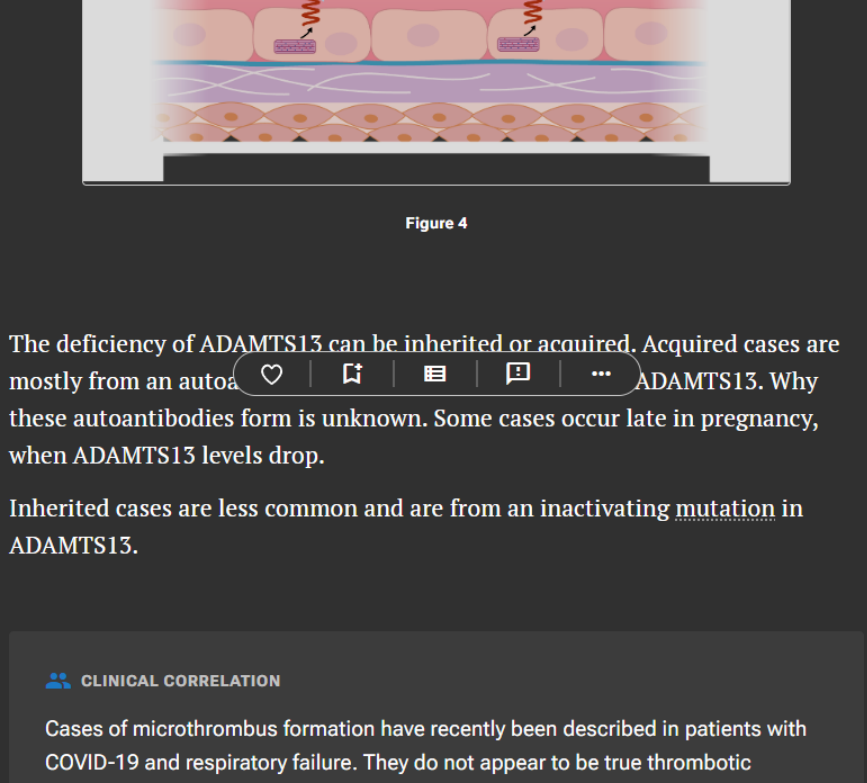
Figure 2

### Thrombotic Thrombocytopenic Purpura

TTP usually presents in previously healthy people. Some have underlying autoimmune diseases like systemic lupus erythematosus. Other risk factors include female sex and conditions such as pregnancy and obesity. The reasons for these risk factors are not known.

**Pathophysiology.** TTP stems from a **deficiency of the enzyme ADAMTS13**, which stands for "a disintegrin and metalloproteinase with a thrombospondin type 1 motif, member 13."

ADAMTS13 normally inhibits thrombosis by cleaving very-high-molecular-weight multimers of von Willebrand factor (vWF) into smaller multimers, which have less tendency to form thrombi. Remember that vWF is a protein that helps platelets a[illegible]ormation of a platelet plug (clot). Normally, when platelets or endothelial cells are activated, they release ultra-large vWF multimers, which are important in forming the clot. In normal patients, ADAMTS13 cleaves these ULvWF multimers into smaller molecules that can then circulate in the blood without causing thrombosis. This prevents excess clotting (Figure 3).


Basement membrane
Endothelium
Subendothelial layer
Blood
Platelet
ADAMTS13
Ultra-large vWF multimers cleaved by ADAMTS13
Circulating plasma vWF multimer

QUIZ

In patients with TTP, a deficiency of ADAMTS13 prevents ULvWF cleavage. Platelets circulating through the blood adhere readily to the large ULvWF multimers, which leads to the **formation of thrombi in small blood vessels** (Figure 4).

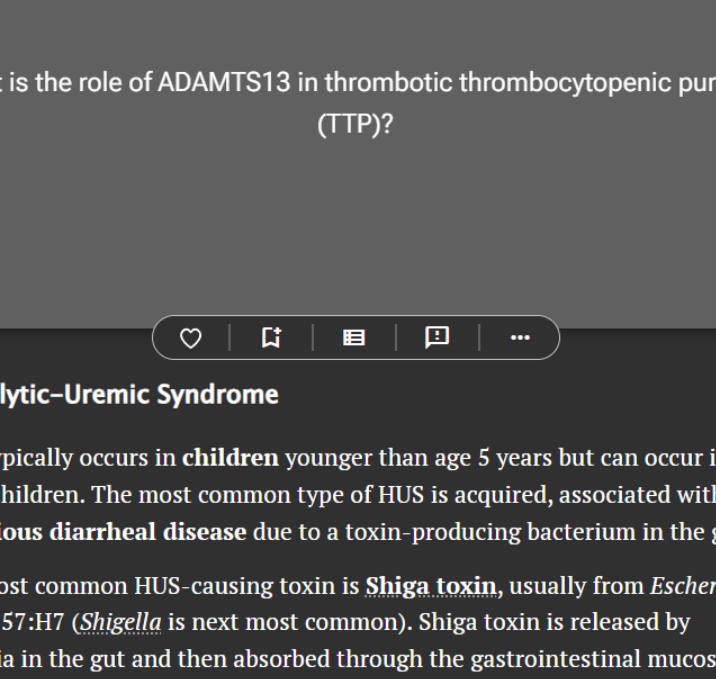


Figure 4

The deficiency of ADAMTS13 can be inherited or acquired. Acquired cases are mostly from an autoa[illegible]ADAMTS13. Why these autoantibodies form is unknown. Some cases occur late in pregnancy, when ADAMTS13 levels drop.

Inherited cases are less common and are from an inactivating mutation in ADAMTS13.

**CLINICAL CORRELATION**

Cases of microthrombus formation have recently been described in patients with COVID-19 and respiratory failure. They do not appear to be true thrombotic thrombocytopenic purpura, however, since ADAMTS13 levels are not decreased, and patients usually have normal platelet counts and lack hemolytic anemia.

What is the role of ADAMTS13 in thrombotic thrombocytopenic purpura (TTP)?

### Hemolytic–Uremic Syndrome

HUS typically occurs in **children** younger than age 5 years but can occur in older children. The most common type of HUS is acquired, associated with an **infectious diarrheal disease** due to a toxin-producing bacterium in the gut.

The most common HUS-causing toxin is **Shiga toxin**, usually from *Escherichia coli* O157:H7 (*Shigella* is next most common). Shiga toxin is released by bacteria in the gut and then absorbed through the gastrointestinal mucosa into the circulation. There, by unclear mechanisms, it alters endothelial cells leading to platelet activation and aggregation.

A rare genetic form of HUS is caused by a defect in complement factors, which leads to unregulated activation of the alternative complement pathway, resulting in endothelial damage and thrombus formation.

What is the toxin and the two bacteria most often responsible for causing hemolytic-uremic syndrome (HUS)?

### Secondary Causes of Thrombotic Microangiopathy

It's important to know some of the important secondary causes of TM because they should always be **considered and excluded** before making a diagnosis of HUS or TTP, which are less common overall. Secondary causes of TM include (Table 1):

**Table 1. Secondary Causes of Thrombotic Microangiopathy**

| | |
|---|---|
| Disseminated Intravascular Coagulation (DIC) | Severe, diffuse activation of both the clotting cascade and formation of thrombi, often in the setting of systemic infection or trauma |
| Hematopoietic stem cell transplantation for treatment of leukemias and lymphomas | From either the associated radiation therapy or chemotherapy |
| HELLP syndrome (hemolysis, elevated liver enzymes, and low platelets) | Some pregnant patients develop MAHA and low platelets, similar to TTP, causing abdominal pain and hypertension; this can progress to placental abruption and organ failure (eg, kidney, lung) |
| Infections | Can lead to direct injury to the vessels; examples include HIV, cytomegalovirus, aspergillosis, and malaria |
| Malignancies | Many are associated with TM, sometimes due to micrometastases to the vessel walls |
| Medications | Calcineurin inhibitors used for transplant rejection (e.g., cyclosporine, tacrolimus), interferon, and vascular endothelial growth factor (VEGF) inhibition cause direct endothelial cell injury; extended-release oxycodone can cause osmotic injury to cells, and quinine can cause vascular [illegible]ng an immune injury |
| Rheumatologic disorders | Systemic sclerosis, systemic lupus erythematosus (SLE), and antiphospholipid syndrome can directly injury blood vessel walls |
| Malignant hypertension (blood pressure >220/100 mm Hg) | Can cause vessel injury, often leading to MAHA, low platelets, and AKI (rising serum creatinine) |

### Pathology

TMs (primary and secondary) are characterized pathologically by MAHA and thrombocytopenia, both due to the formation of thrombi in the vessels.

**Microangiopathic Hemolytic Anemia.** When RBCs travel through small blood vessels, they are sheared by microthrombi, leading to fragmented blood cells called **schistocytes**, which are seen in peripheral blood smears (Figure 5).

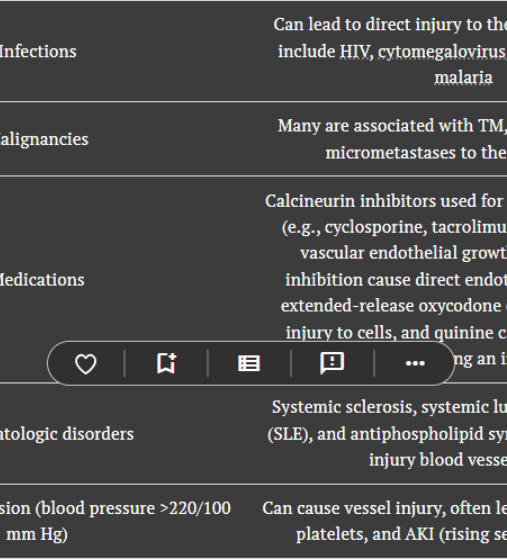


Figure 5

Schistocytes are identifiable by their small size, lack of central pallor, and pointy edges. Schistocytes are specific to MAHA—schistocytes never appear in a normal blood smear, and they are never artifactual. Types of schistocytes include helmet cells (crescent-shaped cells that resemble football helmets) as well as triangulocytes (Figure 6).

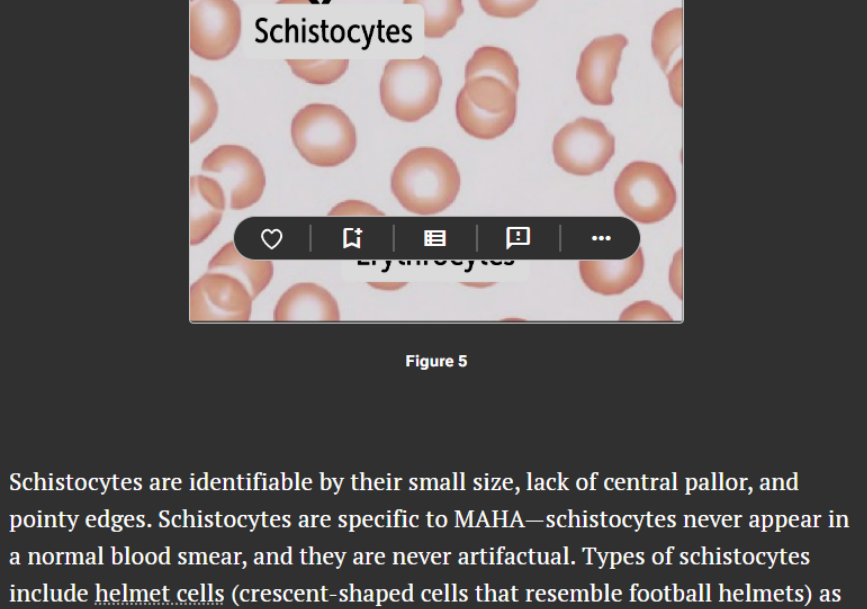


Figure 6

What causes the RBC fragmentation that leads to MAHA? Think of the RBC going through a laser beam (thrombi) that cuts on contact (Figure 7).

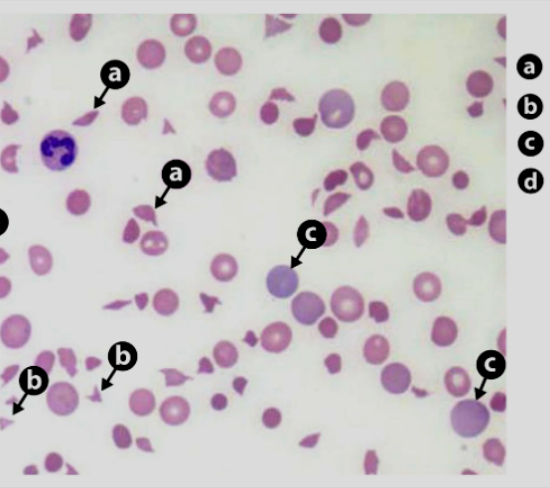


Figure 7

**CLINICAL CORRELATION**

Most, but not all MAHA is due to thrombus formation (TMs). Rarer causes include MAHA from prosthe[illegible]e mechanical shearing of red blood cells (RBCs). Although heart valves are not located in small vessels, they are still considered a cause of MAHA, since the result to the RBCs is the same. Also described is the unusual "march hemolysis," in which people who log many miles either running or marching develop hemolytic anemia due to the fragmentation of RBCs as they pass through the tiny vessels in the soles of the feet.

How does schistocytosis develop in patients with thrombus formation (TM)?

**Thrombocytopenia** results from consumption of platelets as multiple thrombi are formed throughout the vasculature.

Thrombocytopenia has many other causes, including immune destruction.

**Acute Kidney Injury.** In some types of TM (especially HUS and TTP), bacterial toxins and platelet ag[illegible]helium of renal vasculature and obstruct the blood supply to the kidney. Additionally, vomiting leads to volume depletion, which also reduces the GFR.

## How Do We Diagnose Thrombotic Microangiopathies (LO #3)?

To diagnose TM, we first look for signs and symptoms like anemia, hemolysis, and low platelets. We then exclude the secondary causes of thrombotic microangiopathy mentioned earlier (e.g., drugs, hypertensive emergency).

When these secondary causes are excluded, we evaluate for TTP and HUS, which are less common. The distinction between HUS and TTP is not made by lab tests but is usually made by the clinical history and exam.

### Complete Blood Count/Peripheral Blood Smear and Serum Studies

Most patients have anemia with low blood hemoglobin and hematocrit. **Schistocytes** will be seen on the peripheral smear, as will **reticulocytes**, as the bone marrow tries to make more RBC to correct for the hemolysis.

Serum studies will be typical of intravascular hemolytic anemia, showing high unconjugated bilirubin, increased LDH, and low haptoglobin because of the RBC lysis. Note that these studies cannot distinguish between the various causes of TM.

### Coagulation Tests

In TTP, HUS, and most secondary causes of TM, the results on the main coagulation tests (prothrombin time [PT], activated partial thromboplastin time [aPTT], and thro[illegible]

How can these normal coagulation test values be possible when there is such abnormal clotting going on? Recall that the intravascular thrombi result not from activation of the coagulation cascade, but from aggregation of platelets. The thrombi are composed largely of platelets with minimal, if any, contribution from fibrin formation. Coagulation factors are not excessively consumed—so the PT and aPTT are not prolonged.

One exception is DIC, the secondary cause of TM characterized by severe clotting and bleeding throughout the body. In contrast to the other causes, in DIC, the main **coagulation test values are elevated**. This is because the thrombi in DIC result from the overactivation of the coagulation cascade (and are therefore a combination of platelets and fibrin). As coagulation factors are consumed, coagulation studies such as the PT and aPTT quickly become abnormal.

Table 2 summarizes the lab findings in TTP, HUS, and DIC.

**Table 2**

| Condition | aPTT | PT | Thrombin time | Platelets | Schistocyte |
|---|---|---|---|---|---|
| TTP | Normal | Normal | Normal | Decreased | Present |
| Hemolytic-uremic syndrome | Normal | Normal | Normal | Decreased | Present |
| Disseminated intravascular coagulation | Increased | Increased | Increased | Decreased | Present |

Why do coagulation assays have normal results for patients with thrombotic thrombocytopenic purpura (TTP) and hemolytic-uremic syndrome (HUS) but are abnormal in patients with disseminated intravascular coagulation (DIC)?

### Special Testing for Thrombotic Thrombocytopenic Purpura

While clinical exam and routine testing is usually adequate to make a diagnosis of HUS and TTP, in some cases, the cause remains unclear. Special assays can then be done to confirm a diagnosis of TTP from ADAMTS13 deficiency. These suggest immune TTP if the ADAMTS13 activity is under 10% and/or an antibody inhibitor against ADAMTS13 is detected.

**CLINICAL CORRELATION**

As the results of this test usually are not available for several days and treatment for TTP has to start as soon as possible, you cannot wait for these results before deciding whether or not to start treatment.

## How Do We Manage Thrombotic Microangiopathies (LO #4)?

TMs are always treated in consultation with a hematologist. For a secondary cause, the main treatment is removing the causative drug or treating the underlying condition (e.g., hypertension, cancer). Treatment of a primary cause (TTP or HUS) is described next.

### Thrombotic Thrombocytopenic Purpura

TTP is a medical emergency and is managed with **urgent plasmapheresis** (plasma exchange), which helps to clear and replace the defective ADAMTS13. Systemic corticosteroid drugs are used alongside plasmapheresis, and other drugs such as rituximab may be used as well.

### Hemolytic–Uremic Syndrome

HUS treatment is primarily supportive care. Given that most patients present with diarrhea, electrolyte abnormalities, and AKI, this means correcting electrolytes, supporting renal function, and providing IV fluids as needed. Antibiotics are not useful in most cases.

What is the general approach for managing thrombotic [illegible]

**CASE CONNECTION** BACK TO INTRODUCTION ↑

Thinking back to HH, what is the next step in her evaluation? What do you tell her mother about her condition?

Considering her age and constellation of symptoms, you suspect hemolytic-uremic syndrome (HUS). Microangiopathic hemolytic anemia and thrombocytopenia are diagnosed based on the complete blood count and smear. Coagulation assays are done to distinguish between HUS and disseminated intravascular coagulation. They are normal, consistent with HUS. A serum creatinine is elevated. Stool cultures sent on admission return in 24 hours, providing the definitive diagnosis of HUS secondary to a Shiga toxin–producing strain of *Escherichia coli* (*E coli* 0157:H7). HH is admitted to the pediatric intensive care unit.

## Summary

- Thrombotic microangiopathies (TMs) are disorders characterized by diffuse occlusion of small blood vessels by thrombi that lead to hemolytic anemia and low platelets.
- TMs can be prim[illegible] drugs.
- The main primary causes are thrombotic thrombocytopenic purpura (TTP) and hemolytic-uremic syndrome (HUS).
- TTP is rare and seen in adults, whereas HUS is seen mostly in children with a bacterial diarrheal illness.
- Most patients have symptoms of anemia (e.g., fatigue, dizziness, pallor), hemolysis (e.g., jaundice, icterus), and low platelets (e.g., rash, bleeding).
- TTP is characterized by some combination of the pentad of neurologic symptoms, microangiopathic hemolytic anemia, thrombocytopenia, fever, and acute kidney injury (AKI).
- HUS is similar to TTP but lacks neurologic findings.
- In all TMs, schistocytes and reduced platelets are seen on the peripheral blood smear.
- TTP is caused by a deficiency of ADAMTS13 that cleaves large von Willebrand factor (vWF) multimers into smaller components to prevent platelet aggregation.
- HUS is usually caused by Shiga toxin released by *E coli* O157:H7 or *Shigella*.
- Secondary causes are often due to direct injury to endothelial cells (e.g., by hypertension, drugs, malignancy).
- Peripheral blood smear and coagulation tests cannot distinguish between most causes, including TTP vs HUS, so the distinction is made by history and physical exam.
- Schistocytes, anemia, and thrombocytopenia are present in most patients.
- Coagulation tests are all normal in most causes of TM except disseminated intravascular coagulation (DIC), in which they are instead prolonged.
- TTP is managed with plasmapheresis and glucocorticoids.
- HUS management is primarily supportive care including fluids.
- Secondary causes are managed by treating the underlying disease or removing the cau[illegible]

## Review Questions

**1. Which of the following blood smear findings would most likely be seen in hemolytic anemia due to any cause?**

- **A.** ○ Bite cells
- **B.** ○ Crescent-shaped RBCs
- **C.** ● Reticulocytes
- **D.** ○ Schistocytes
- **E.** ○ Triangulocytes

**Hide Explanation**

The correct answer is reticulocytes (C). The presence of reticulocytes in the peripheral blood is seen in all forms of hemolytic anemia, as long as the bone marrow is healthy. It indicates an effort on the part of the bone marrow to compensate for hemolysis by making more red blood cells (RBCs), releasing immature RBCs into the circulation as part of the increased erythropoiesis. The other responses are seen only in specific types of hemolytic anemia. Bite cells (A) are typically seen in patients with glucose-6-phosphate dehydrogenase deficiency. Crescent-shaped RBCs (B), also called sickle cells, are characteristic of sickle cell anemia. Schistocytes (D) and triangulocytes (E) are the result of RBC fragmentation and their presence o[illegible]angiopathic hemolytic anemia (MAHA).

**2. Which of the following best describes the pathophysiology of anemia in thrombotic microangiopathy?**

- **A.** ○ Autoantibodies attacking RBCs
- **B.** ○ Blunt trauma leading to massive bleeding
- **C.** ● Fibrin strands within blood vessels leading to fragmentation of RBCs
- **D.** ○ Genetic mutation leading to malformation of globin protein
- **E.** ○ Vitamin deficiency leading to impaired development of RBCs

**Hide Explanation**

The correct answer is fibrin strands within blood vessels leading to fragmentation of RBCs (C). Schistocytes in microangiopathic hemolytic anemia are caused by trauma to RBCs, either from turbulent flow in obstructed blood vessels or by mechanical prosthetic heart valves. Autoantibodies attacking RBCs (A) are seen in autoimmune hemolytic anemia, and blunt trauma leading to massive bleeding (B) can cause anemia through blood loss. Genetic mutation leading to malformation of globin protein (D) is seen in thalassemia. Vitamin deficiency leading to impaired development of RBCs (E) can be seen in folate, $B_{12}$, and $B_6$ deficiencies, but these are a defect in erythropoiesis, not seen in thrombotic microangiopathies.

**3. A 5-year-old male has diarrhea and vomiting after attending a picnic. His exam shows pallor, yellow sclera, limb petechiae, sunken eyes, and irritability. The serum indirect bilirubin is elevated. Which of the following is most likely on his peripheral blood smear?**

- **A.** ○ Blasts
- **B.** ○ Howell-Jolly bodies
- **C.** ● Schistocytes
- **D.** ○ Target cells
- **E.** ○ Thrombocytosis

**Hide Explanation**

The correct answer is schistocytes (C). The child has hemolytic-uremic syndrome (HUS) characterized by diarrheal illness followed by low platelets (e.g., petechiae), hemolytic anemia (e.g., icterus, pallor, high bilirubin), and dehydration (e.g., sunken eyes). Microangiopathic hemolytic anemia is seen in HUS, with increased bilirubin, jaundice, and pallor, and schistocytes would be seen on the blood smear. The other structures would not be associated with diarrheal illness or hemolysis. Blasts (A) are immature white blood cells (WBCs) seen in patients with leukemia or myeloproliferative disorders. Howell-Jolly bodies (B) are nuclear remnants of red blood cells (RBCs). They are expelled during maturation but may persist in cases of asplenia, megaloblas[illegible]thers. Target cells (D) are an abnormal morphology of RBCs in which a dark ring is seen surrounding a clear center and are most commonly seen in thalassemia. Thrombocytosis (E) is defined as an increased platelet count, but HUS would instead cause decreased platelets.

**4. An adult female is brought into the emergency department because of increasing confusion and bleeding gums for 2 days. The physical exam shows a temperature of 38.6°C (101.5°F), disorientation to time and place, and petechiae and purpura on both legs. Lab exam shows reduced platelets, anemia, and schistocytes. What is the most appropriate treatment for this patient's condition?**

- **A.** ○ Broad spectrum antibiotics
- **B.** ○ Corticosteroids
- **C.** ● Plasma exchange
- **D.** ○ Supportive care

**Hide Explanation**

The correct answer is plasma exchange (C). This patient is presenting with TTP, and urgent plasma exchange is the key treatment. Corticosteroids (B) may also be used, but plasma exchange is key because it removes the inhibitory autoantibody and the abnormally large VWF multimers. Broad spectrum antibiotics (A) would not be useful in this condition, give[illegible]pportive care (D) is used for the treatment of HUS, not TTP.

**5. Which of the following lab tests best differentiates between primary thrombotic microangiopathies and disseminated intravascular coagulation (DIC)?**

- **A.** ● aPTT and PT
- **B.** ○ D-dimers
- **C.** ○ Fibrin degradation products
- **D.** ○ Peripheral blood smear
- **E.** ○ Platelet count

**Hide Explanation**

The correct answer is aPTT and PT (A). In the primary (and most secondary) TMs, the main coagulation assays are all normal, whereas they are increased in disseminated intravascular coagulation (DIC) because coagulation factors are consumed in the disease process. D-dimers (B) and fibrin degradation products (C) are elevated in most forms of thrombosis, so are not helpful in this distinction. Any form of thrombotic microangiopathy will show schistocytes and reduced platelets on peripheral blood smear (D) and have a decrease in platelet count (E).

**How would you rate this Brick?**

## References

- Hoffman R, Benz EJ, Silberstein L, et al. *Hematology: Basic Principles and Practice*. 8th ed. Elsevier; 2023.

## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☑ ~~Differentiate the clinical presentation of patients with thrombotic microangiopathies (TMs).~~
- 02 ☑ ~~Explain the pathophysiology of TMs.~~
- 03 ☑ ~~Interpret clinical findings to properly diagnose TMs.~~
- 04 ☑ ~~Outline the management plan of TMs.~~

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Objective Confidence: 4/4